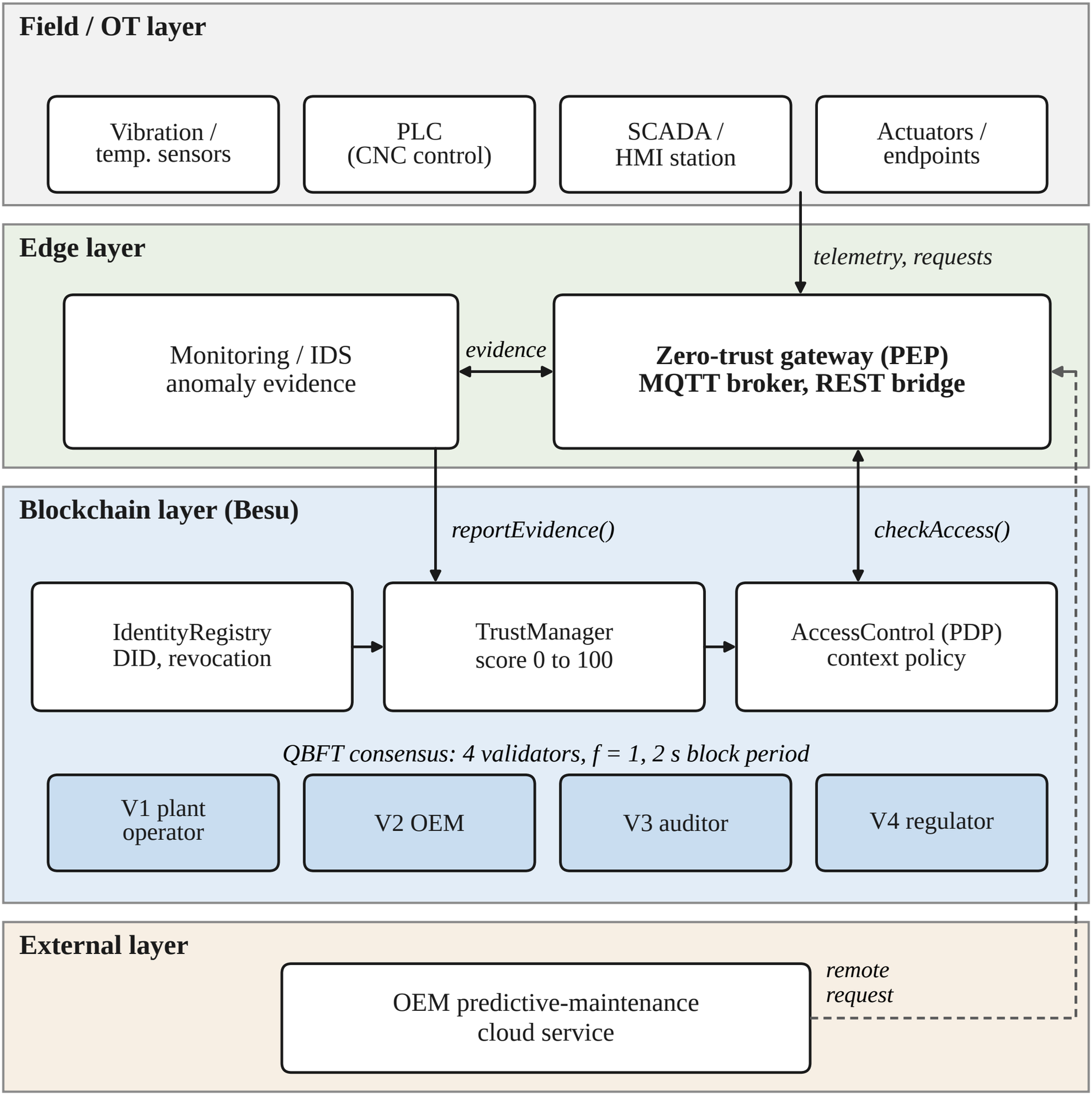
Field / OT layer
Vibration / temp. sensors
PLC (CNC control)
SCADA / HMI station
Actuators / endpoints
telemetry, requests
Edge layer
Monitoring / IDS anomaly evidence
evidence
Zero-trust gateway (PEP) MQTT broker, REST bridge
Blockchain layer (Besu)
reportEvidence()
checkAccess()
IdentityRegistry DID, revocation
TrustManager score 0 to 100
AccessControl (PDP) context policy
QBFT consensus: 4 validators, f = 1, 2 s block period
V1 plant operator
V2 OEM
V3 auditor
V4 regulator
External layer
OEM predictive-maintenance cloud service
remote request